

PROBLEM-SOLUTION FIT

Date	04 July 2026
Team ID	2026-4860
Project Name	EduGenie Google Gemini Powered Learning Assistant
Maximum Marks	5 Marks

PROBLEM-SOLUTION FIT CANVAS:

Use the framework below to ensure your proposed solution accurately addresses the root causes, behaviors, and limitations of your target audience. Fill out the canvas in the respective boxes just like the original visual template.

1. CUSTOMER SEGMENT(S) [CS] - Students (School & College level) - Self-learners exploring new tech/skills (e.g., SQL, Math) - Educators looking for automated quiz/summary tools	6. CUSTOMER LIMITATIONS [CL] Budget: Can't afford expensive premium AI subscriptions Devices: Resource-constrained local machines (e.g., base Mac M1, old laptops) Internet: Continuous high-speed data dependency for heavy cloud models	5. AVAILABLE SOLUTIONS PROS & CONS [AS] ChatGPT / Premium Bots: - Pros: Powerful, wide knowledge base. - Cons: High cost, generic responses, requires heavy internet. Traditional E-Learning Platforms: - Pros: Structured paths. - Cons: Static content, no instant interactive Q&A.
2. PROBLEMS / PAINS + ITS FREQUENCY [PR] - Difficulty understanding complex topics instantly (Frequency: Daily during study) - Time-consuming manual text summarization and lack of self-evaluation quizzes (Frequency: High during exams)	9. PROBLEM ROOT / CAUSE [RC] - Lack of 24/7 personalized tutors or immediate doubt-clearing sources - Static educational content that does not adapt to individual learning speeds - Heavy AI models are slow or inaccessible offline/locally	7. BEHAVIOR + ITS INTENSITY [BE] - Students open multiple tabs, searching across Google/YouTube for simple explanations. Intensity: Very High/Anxious during exam prep or when stuck on a coding/math logic.
3. TRIGGERS TO ACT [TR] - Getting stuck on a complex question (e.g., "Which is the largest ocean?") - Needing an urgent quick test/quiz before an exam - Starting a completely new topic (e.g., SQL) and not knowing the roadmap	10. YOUR SOLUTION [SL] EduGenie: A lightweight FastAPI-based AI Assistant combining local & cloud models. Key Deliverables: - Intelligent Question Answering & Text Summarization - Instant customized Quiz Generation - Structured Roadmap/Learning Paths (Gemini 1.5 Pro & LaMini-Flan-T5) - Optimized architecture to run	8. CHANNELS OF BEHAVIOR [CH] ONLINE: Web-based responsive UI (HTML+CSS+Jinja2), GitHub repositories, Educational forums.

flawlessly on resource-constrained hardware (Mac M1).

4. EMOTIONS BEFORE / AFTER [EM]

- **Before:** Overwhelmed, confused, stressed by heavy study materials.
- **After:** Confident, clear, motivated with structured roadmaps and quick concept clarity.

OFFLINE:

Local deployment on student laptops, college computer labs, offline study groups.